

MEL1010 Engineering Mechanics

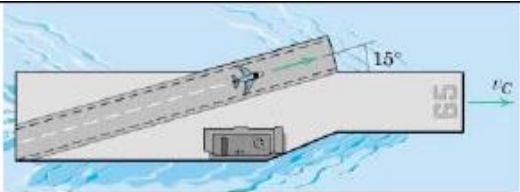
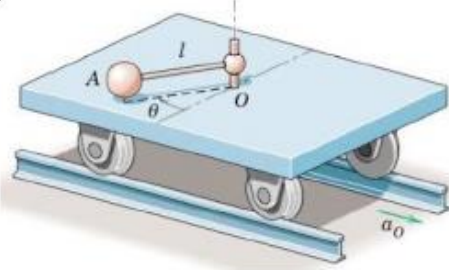
Tutorial 7: Relative Motion, System of Particle and Rigid body Kinematics-

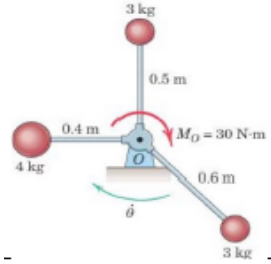
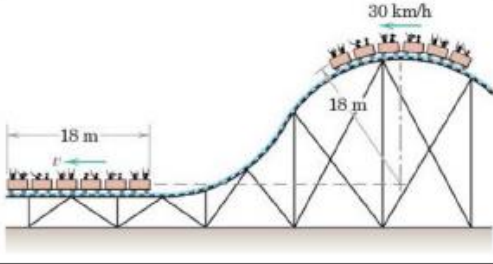
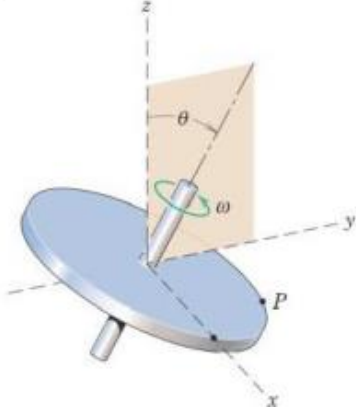
Absolute motion

Thursday, 17th October

Answer all the following questions. Mention valid assumptions wherever applicable

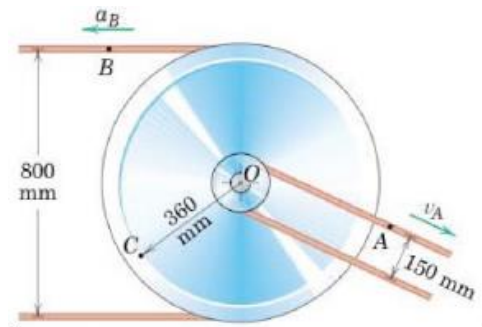
1-2: Relative motion, 3-4: System of Particles, 5-6: Rotation, 7-9: Absolute Motion

1.	The launch catapult of the aircraft carrier gives the 7-Mg jet airplane a constant acceleration and launches the airplane in a distance of 100 m measured along the angled takeoff ramp. The carrier is moving at a steady speed $v_c = 16$ m/s. If an absolute aircraft speed of 90 m/s is desired for takeoff, determine the net force F supplied by the catapult and the aircraft engines. Ans. $F = 194.0$ kN	
2.	The ball A of mass 10 kg is attached to the light rod of length $l = 0.8$ m. The mass of carriage alone is 250 kg, and it moves with an acceleration a_0 as shown. If $\dot{\theta} = 3$ rad/s when $\theta = 90^\circ$, find the kinetic energy T of the system if the carriage has a velocity of 0.8 m/s (a) in the direction of a_0 and (b) in the direction opposite to a_0. Treat the ball as a particle. Ans. $T = 112$ J	

3.	The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20$ rad/s. If a couple $M_0 = 30 \text{ N} \cdot \text{m}$ is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$. <i>Ans.</i> $\dot{\theta}' = 80.7 \text{ rad/s}$	
4.	The cars of a roller-coaster ride have a speed of 30 km/h as they pass over the top of the circular track. Neglect any friction and calculate their speed v when they reach the horizontal bottom position. At the top position, the radius of the circular path of their mass centers is 18 m, and all six cars have the same mass. <i>Ans.</i> $v = 72.7 \text{ km/h}$	
5.	The circular disk rotates with a constant angular velocity $\omega = 40 \text{ rad/sec}$ about its axis, which is inclined in the y-z plane at the angle $\theta = \tan^{-1} \frac{3}{4}$. Determine the vector expressions for the velocity and acceleration of point P, whose position vector at the instant shown is $\mathbf{r} = 15\mathbf{i} + 16\mathbf{j} - 12\mathbf{k}$ in. (Check the magnitude of your results from the scalar values $v = r\omega$ and $a = r\omega^2$) $\mathbf{v} = 40(-20\mathbf{i} + 12\mathbf{j} - 9\mathbf{k}) \text{ in/s}$ $\mathbf{a} = 1600(-15\mathbf{i} - 16\mathbf{j} + 12\mathbf{k}) \text{ in/s}^2$	

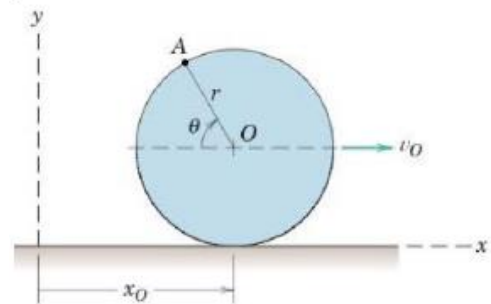
6. The two V-belt pulleys from an integral unit and rotate about the fixed axis at O . At a certain instant, point A on the belt of the smaller pulley has a velocity $v_a = 15 \text{ m/s}$, and point B on the belt of larger pulley has an acceleration $a_B = 45 \text{ m/s}^2$ as shown. For this instant determine the magnitude of the acceleration a_c of point C and sketch the vector in your solution.

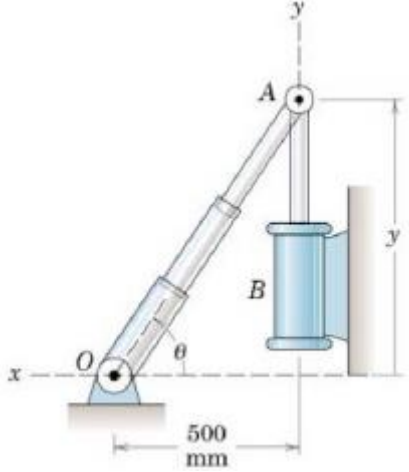
Ans. $a_c = 149.9 \text{ m/s}^2$



7. The wheel of radius r rolls without slipping, and its centre O has a constant velocity v_0 to the right. Determine expressions for the magnitudes of the velocity v and acceleration a of point A on the rim by differentiating its x - and y -coordinates. Represent your results graphically as vectors on your sketch and show that v is the vector sum of two vectors, each of which has a magnitude v_0

Ans. $v = v_0 \sqrt{2(1 + \sin \theta)}$, $a = \frac{v_0^2}{r}$ towards O



8.	The telescoping link is hinged at O, and its end A is given a constant upward velocity of 200 mm/s by the piston rod of the fixed hydraulic cylinder B. Calculate the angular velocity $\dot{\theta}$ and the angular acceleration $\ddot{\theta}$ of link OA for the instant when $y = 600$ mm. <i>Ans.</i> $\dot{\theta} = 0.1639$ rad/s $\ddot{\theta} = -0.0645$ rad/s²	
9.	Determine the acceleration of the shaft B for $\theta = 60^\circ$. If the crank OA has an angular acceleration $\ddot{\theta} = 8$ rad/s² and an angular velocity $\dot{\theta} = 4$ rad/s at this position. The spring maintains contact between the roller and the surface of the plunger. <i>Ans.</i> $a_B = 789$ mm/s²	